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HighGen—Comparative genomics resources for cannabis

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Reference-guided whole-genome alignment pipelines are tuned for cross-species vertebrate panels. With **HighGen**, we adapted them in Galaxy across **217** ***Cannabis sativa*** **assemblies** anchored on cs10 through standardized repeat masking, empirical substitution scoring, chunked extension, and streaming 64-bit dynamic programming. We pair the alignment with 43-genome Panagram *k*-mer bitmaps, a TOGA2 orthology pilot, and a 78-family phenotype ledger.

Galaxy PROJECT

228-stanza (219-assembly) Hub
Attach depot.galaxyproject.org/highgen/hub.txt in the UCSC Genome Browser (Track Hubs → My Hubs).

A Four cross-species bottlenecks

In *Cannabis sativa*, interspersed repeats (~64–67% LTR retrotransposons) surround dense tandem metabolic arrays (Grassa et al. 2021, Lynch et al. 2025). Projecting 216 query assemblies onto the linear cs10 (CBDRx) reference coordinate grid confronts four vertebrate-default obstacles:

1 Masking bias
Arrived soft-masking spans 0.00–80.86% across 218 assemblies, and KegAlign (Gulhan et al. 2025), in our measurement, neither seeds nor extends through lowercase bases

2 Cross-species matrix
HOXD70 (Chiaromonte et al. 2002) targets below 70% identity, which in *Cannabis* is 49.7% chance noise (Ts/Tv = 0.872 vs. 0.74 floor)

3 Superlinear scaling
Whole-chr LASTZ (Harris 2007) scales as $O(n^{1.9})$, while 217-way MULTIZ (Blanchette et al. 2004) overflows 32-bit scores and buffers MAFs in RAM

4 Macro-syntenic nets
Human-mouse netFilter -syn (Kent et al. 2003, maxFar = 200 kb) prunes rearranged tandem gene clusters

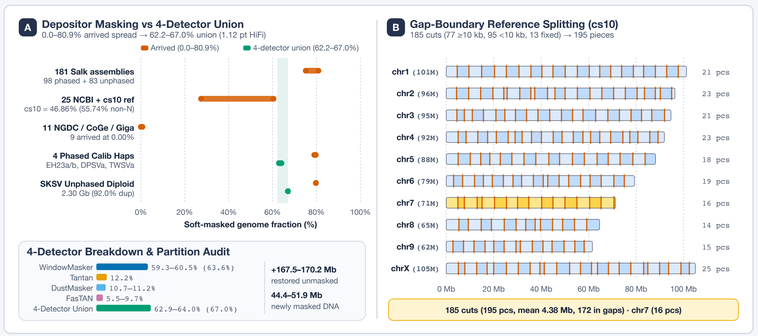


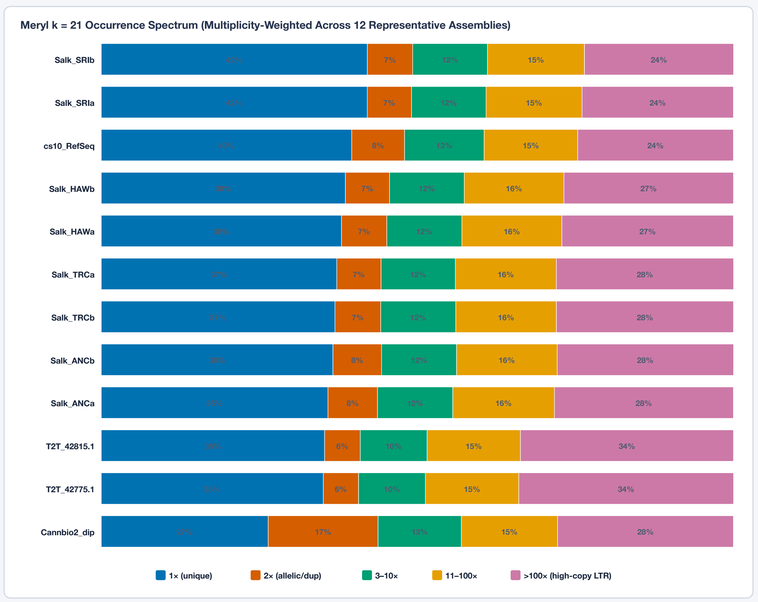
Figure 1 | Masking convergence and gap-boundary reference splitting. Panel **A**: stripping depositor masks across the 5-genome calibration panel and 5-genome intake pilot collapses a 0.0–80.9% intake spread into a 62.2–67.0% 4-detector union. Panel **B**: across cs10, 185 chromosomal cuts (77 at N-runs ≥ 10 kb, 95 at N-runs under 10 kb, and 13 fixed 5 Mb cuts in wfi_cs10_splits.bed) partition the assembly into 195 ~5 Mb pieces plus 26 unplaced scaffolds, mitochondrion NC_029855.1, and small.00. mafSplit divides straddling blocks without base loss, and maf_concat rejoins the pieces.

B Standardized 4-detector repeat masking

We curated 221 assembly records—217 non-redundant in MULTIZ (cs10 plus 216 queries across 164 prefix groups and 156 plants, scored with rosales_odb12 BUSCO, Lynch et al. 2025, Pike et al. 2025, 2026). To remove submitter ascertainment bias, our brc-tools UDT wrappers strip arrived masks and unite four *de novo* detectors (Morgulis et al. 2006, Frith 2011):

fasta-uppercase → WindowMasker DustMasker Tantan FasTAN → bedtools merge

Assembly / Cohort	Arrived Mask	4-Detector Union	Partition Audit and Compute
181 Salk assemblies (98 phased + 83 unphased)	75.12–80.86%	**62.19–67.04%** (5-genome intake pilot: 22.61 → 4.85 pt spread)	Depositors over-masked ~22% diverged unmasked DNA
25 NCBI GenBank + cs10	27.42–60.43% (cs10 46.86% total, 55.74% non-N)		Leaves 44.26% of cs10 non-N bases open to seeding
11 NGDC / CoGe / GigaDB	0.00–0.68% (9 kept at 0.00%)		Prevents unmasked queries from seeding 100% of repeats
4 Phased Calibration Haps (EH23a/b, DPSVa, TWSVa)	**79.00–79.61%**	**62.90–64.02%** (1.12-pt window)	**WM 59.30–60.53%, Dust 10.7–11.2%, Tantan 12.2%, FasTAN 5.5–9.7% · 3.64–3.99 CPU-h/hap**
Diploid SKSV (2.30 Gb, 92.0% BUSCO dup)	79.71%	**66.95%** (4.05-pt panel window, WM 63.60%)	432.1 Mb restored, 139.1 Mb newly masked · 10.34 CPU-h



12-genome Meryl *k* = 21 occurrence spectrum: Phased haplotypes (cs10, 8 Pike et al. 2026 ONT, 2 IMPD T2T) carry 35–42% 1× and 6–8% 2× multiplicity-weighted *k*-mers, whereas uncollapsed diploid Cannbio-2 (Braich et al. 2020) carries 27% 1× and 17% 2×.

Across phased haplotypes, the union restores 167.5–170.2 Mb of depositor-masked DNA while newly masking 44.4–51.9 Mb—a net unmasked gain of +115.9 to 123.2 Mb per haplotype with exact lowercase FASTA ↔ 0-based BED parity.

C Empirical intra-species substitution matrix

We adapted the Chiaromonte et al. (2002) log-odds protocol across two phased haplotype pairs (EH23a × EH23b and DPSVa × TWSVa, 95.3% rosales_odb12 BUSCO completeness, 1.8–2.0% duplication). We inverted the training window to [85, 95)% identity, excised Salk GFF3 protein-coding CDS (–7.3% pre-chain segments), and enforced single-chain colinearity (--max-chains 1). This filter retained 54.5% of orthologous vs. 6.6% median off-chromosome paralogous bases (**8.2-fold**) on Cb2 × cs10 and **18.0-fold** genome-wide on EH23a × EH23b.

82.76 Mb
Physical [85, 95)% pairs (331.05M 4×-symmetrized, Ts/Tv = 2.119, *H* = 1.4140 bits, either pair alone within 1 unit)

10/10 Arms
Wins vs. HOXD70 (chr8 re-run after OOM, 9/10 loci $p = 0.0107$ vs. $\alpha = 0.0083$) and 67/70 windows (95.7% [91.4, 100%])

5,033×
axtChain shuffled-null suppression (EH23a chr1 × *Morus*) + scaled 682/51 gaps (70.6% single coverage)

We normalized scores to a maximum match of 100 (+80 A/T, +100 C/G, –92 transitions, –191 A-C/G-T, –197 C-G, –198 A-T transversions). Across 40 false-match budgets, the matrix beat unit ±100 on **10/10 arms**, **10/10 loci**, and **62/70 windows** (88.6%, CI 83.5–94.6%). It also beat the unchained matrix on **55/70 windows** (78.6%, CI 70.3–85.5%, 9/10 loci), with 0.02–0.11% arm margins and crossing budget curves on every locus.

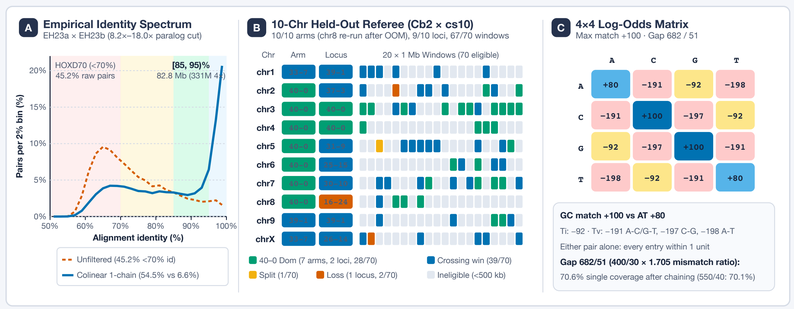


Figure 2 | Intra-species matrix calibration and 10-chromosome × 20-window held-out referee. Panel **A**: empirical identity spectra and single-chain colinearity separation. Panel **B**: antisymmetric corner-threshold wins across 10 held-out Cb2 × cs10 chromosomes and 70 eligible 1 Mb windows. Panel **C**: symmetric 4×4 log-odds weights with 682/51 gap scaling.

D Collection-native LASTZ and streaming MULTIZ 11.4

- **Growler LASTZ (growler_ucsc_pair, LASTZ 1.04.52/1.04.60):** Couples KegAlign v0.3.4 GPU collection seeding (Gulhan et al. 2025) with indexed .2bit archives, compact LAV, and piped PSL chaining (Harris 2007, Kent et al. 2003). Batching anchor files into ≤50,000 -segment chunks cuts EH23a × EH23b gapped extension **7.8-fold** (389.0 → 49.8 slot-h, 1.88 h wall-clock, median 28 min/job, under 3.97 GB RAM). Output matches **99.99%** of UCSC's hg38 chr21 × panTro6 chr21 all.chain and rbest.chain pairs and yields byte-identical doRecipBest chains across 5 test queries.
- **Storage and 218-run production cohort:** Purging transient intermediate collections after workflow completion cut total Galaxy history storage across all 218 runs (54,554 jobs, 10,325 job-hours) from **14.96 TB to 6.50 TB (–56.5%,** 8.45 TB saved). Syntenic nets cover a median **65.2%** of cs10 non-N bases across 216 queries (range 12.2–85.1%, vs. **70.0%** and 29.3–88.1% in rbest, with 40.0% diploid vs. 54.0% phased median query-side coverage).
- **Streaming 64-bit column-count MULTIZ 11.3/11.4:** In v11.3 (multiz#19, Blanchette et al. 2004), coordinate-ordered streaming cursors replaced in-memory MAF buffering (v11.2) and int64_t accumulators fixed 32-bit overflow across 29 deep blocks on NC_044376.1.02 (24 with scores $\geq 2^{31}$). Peak memory fell **51.7-fold** on hg38 5-way chrX (677 → 13 MiB, 217-way worker RAM ≤14.1 MB). In v11.4 (multiz#28), factoring mz_yama.c DP loops through column nucleotide counts (c_A, c_C, c_G, c_T, c_N) sped up a deep 216-genome block merge **33.8-fold** (981 → 29 s vs. v11.3). Whole-piece runtime on a 4.1 Mb segment dropped **6.5-fold** (7.3 → 1.1 h, 82 chr pieces merged under v11.3 and 113 + scaffolds under v11.4) with byte-identical output.

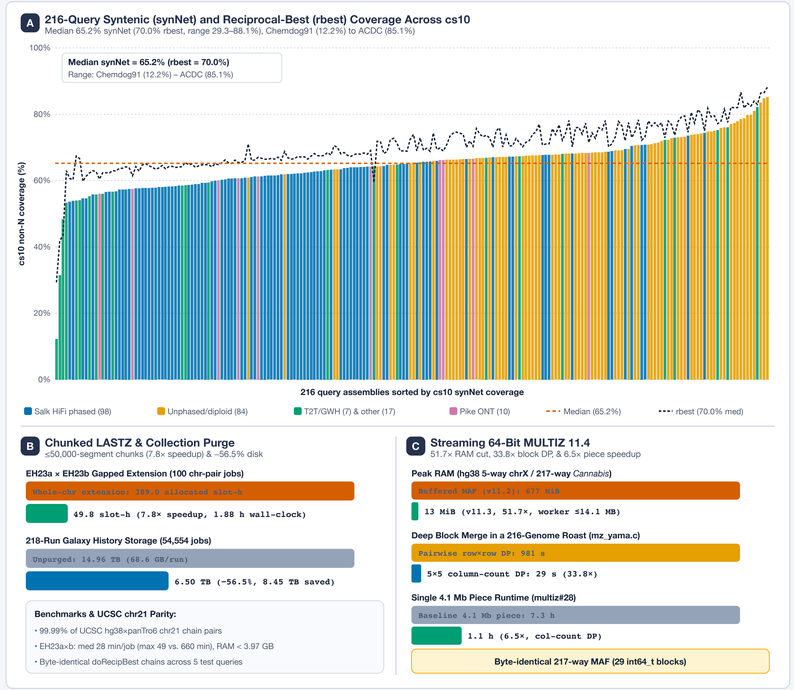


Figure 3 | 216-query syntenic coverage waterfall and scalability benchmarks. Panel **A**: sorted synNet (median 65.2%, range 12.2–85.1%) and rbest 1-to-1 (median 70.0%, range 29.3–88.1%) coverage across 216 queries by assembly class. Panel **B**: chunked LASTZ speedup, hg38 chr21 parity, and 218-run storage purge. Panel **C**: MULTIZ v11.3/v11.4 streaming RAM reduction (51.7×) and column-count DP speedup (33.8× block, 6.5× piece).

E Tandem metabolic clusters and syntenic net limits

- **Genome-wide tandem census and permutation audit:** Under MCScanX tandem criteria (Wang et al. 2012) adapted to stricter BLASTp thresholds ($E \leq 10^{-10}$, top 10 hits), **3,035** of cs10's 25,296 RefSeq protein-coding genes reside in 1,208 tandem arrays ($K = 0$) and **5,123 (20.3%)** in 1,796 clusters ($K \leq 10$). These clusters include **41 terpene synthases (TPS)** (31 in 9 clusters + 10 singletons) and **35 cannabinoid oxidocyclases (BBE-like)** (29 in 9 clusters + 6 singletons including CBDAS). Across 20,000 cluster-level permutations, nominal TPS low-identity enrichment (37.0% vs. 11.0% controls, $p = 0.022$) does not survive family-wise correction ($p_{FWER} = 0.12$, threshold-free $p = 0.40$, BBE-like 2.8%, $p = 0.59$).
- **Functional CBDAS vs. B-locus pseudogenes:** At cs10 functional CBDAS (LOC115697762, chr7:30.98 Mb, Grassa et al. 2021), our 217-way alignment separates **79 assemblies** (59 plants, including 56 Lynch full-length CBDAS) at ≥98.5% identity and **6 assemblies** below 93.0% (≥50% CDS) from **77 assemblies** (67 plants) at 93.0–98.5%. In that middle band, all 63 Lynch-called contigs (62 THC-type, 1 both, 14 uncalled, Lynch et al. 2025) pair cs10 CBDAS with the ~94% B-locus pseudogene near THCAS.
- **netFilter -syn pruning vs. paralog reuse:** Human-mouse cutoffs (maxFar = 200 kb, plus minTopScore = 300,000 for 14.5% of losses) pruned **4.7% of TPS** (418/8,856: 4.2% clustered, 6.4% singleton) and **10.7% of BBE-like** (807/7,560: 7.5% clustered, 26.2% singleton vs. 5.2%/13.1% controls). In synNet, this dropped 54 CBDAS assemblies (under 50% CDS), including Finola's intact 99.4% transposed copy (CM011610.1:21.84 Mb, 26.8 Mb from its enclosing gap). Removing netFilter -syn leaves **46.4%** of recovered pairs (569/1,225) overlapping another paralog's interval. We therefore provide complementary rbest nets alongside TOGA2 and TOAST/BOAST orthology (Hou et al. 2006, 2009).

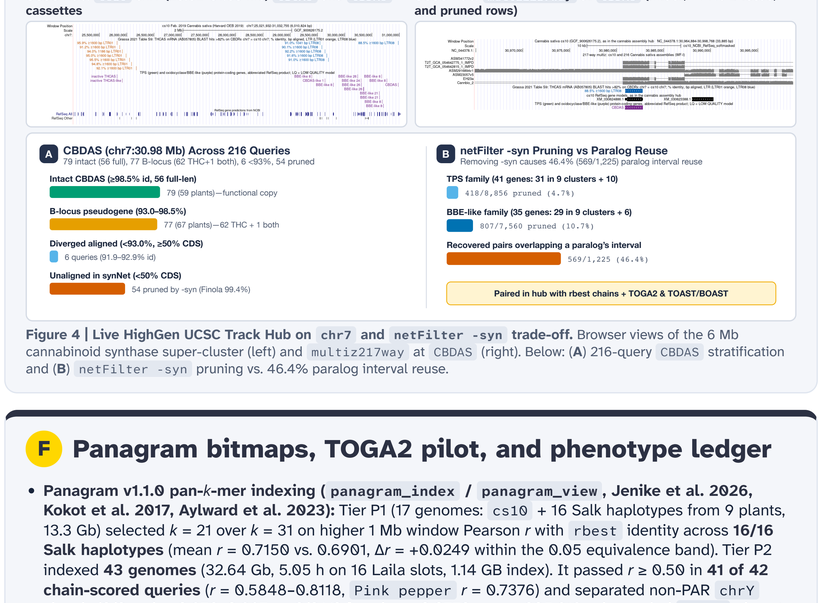


Figure 4 | Live HighGen UCSC Track Hub on chr7 and netFilter -syn trade-off. Browser views of the 6 Mb cannabinoid synthase super-cluster (left) and multiz217way at CBDAS (right). Below: (**A**) 216-query CBDAS stratification and (**B**) netFilter -syn pruning vs. 46.4% paralog interval reuse.

F Panagram bitmaps, TOGA2 pilot, and phenotype ledger

- **Panagram v1.1.0 pan-*k*-mer indexing (panagram_index / panagram_view, Jenike et al. 2026, Kokot et al. 2017, Aylward et al. 2023):** Tier P1 (17 genomes: cs10 + 16 Salk haplotypes from 9 plants, 13.3 Gb) selected $k = 21$ over $k = 31$ on higher 1 Mb window Pearson r with rbest identity across **16/16 Salk haplotypes** (mean $r = 0.7150$ vs. 0.6901, $\Delta r = +0.0249$ within the 0.05 equivalence band). Tier P2 indexed **43 genomes** (32.64 Gb, 5.05 h on 16 Laila slots, 1.14 GB index). It passed $r \geq 0.50$ in **41 of 42 chain-scored queries** ($r = 0.5848$–0.8118, Pink pepper $r = 0.7376$) and separated non-PAR chrY cleanly (3/3 males 0.976–0.987 vs. 3/3 V3 females ≤ 0.0773 across 205,619 subsampled AH3Ma -absent positions). It flagged 2 Guppy 5 SUP trio-binned ONT haplotypes on G3 count tolerance (Punto Rojo –10.4%, Cherry Pie –10.8%, Pike et al. 2025). Only one Sri Lankan ONT haplotype fell below 0.50 (ASM5191173v1, $r = 0.4763$, Merqury QV 24.01/24.11, Pike et al. 2026), for reasons still under investigation.
- **TOGA2 v2.0.9 orthology pilot (Kirilenko et al. 2023, Malovichko et al. 2026):** On cs10, we stratified all **25,261** nuclear protein-coding genes (33,639 transcripts, excluding mitochondrion NC_029855.1) into **24,503 standard** and **758 reference-anomaly** genes alongside 221 inventoried all.chain datasets (1.45 TB). On its zero-divergence identity-chain control (C-floor), TOGA2 called **25 of 24,503 lost/uncertain (0.10%)**—1 L + 24 UL against a ≤ 0.5% bar. On EH23a, the production, loose, and RepeatFiller -filled-and-cleaned chains called **2,077 (8.48%)**, **1,778 (7.26%)**, and **1,660 (6.77%)** standard genes L/UL. Filled and production arms disagree on 713 of 2,161 L/UL/M genes (33.0%), while production yields the fewest multi-copy false presences (4,306 standard genes vs. 5,058 loose and 5,245 filled). Because our top-ranked native-model check (B6) is still pending, the pilot selects no arm and reports no gene loss.
- **Pedigree-verified phenotype ledger:** We joined 40 sources across 215 assemblies—30 publications (29 papers, including 2 supplements, plus 988 Lynch et al. 2025 Figshare rows across 190 assemblies from 43 audited extractions in passed.tsv) and 10 NCBI BioSamples. This yielded **1,341 trait records** across **154 plants in 78 pedigree families** (69 plants in the largest Salk family, 71 singletons). Under strict measured / reported filtering, 0 of 14 traits pass our ≥30-family, ≥2-class (≥5 families/class) association screen. Varin spans 30 families (37/38 non-WT plants in HO40 across 3 classes), while Chemotype reaches 30 families with 2 classes ≥5 only if Lynch's HPLC methods sentence covers its table column.

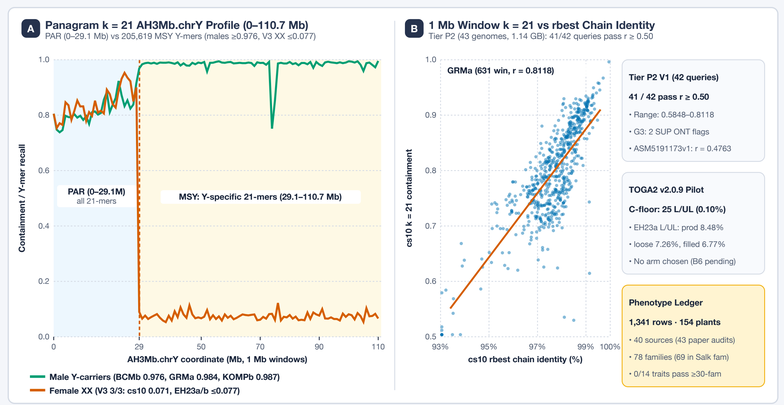


Figure 5 | Alignment-free Paranagram chrY profile and rbest correlation. Panel **A**: chromosome-wide AH3Mb.chrY (0–110.7 Mb) $k = 21$ profile of PAR all-21-mer containment (0–29.08 Mb) and MSY male-specific 21-mer recall across 205,619 subsampled positions. Panel **B**: all 631 eligible 1 Mb windows on GRMa ($r = 0.8118$) with Tier P2, TOGA2, and phenotype scorecards.

Open tools, workflows, and 228-stanza (219-assembly) browser hub

4-detector masking → brc-tools **GPU seeding + chunked LASTZ → growler_lastz_lav** **217-way streaming TBA → MULTIZ 11.4** **Pan-*k*-mer + orthology → Panagram and TOGA2**

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HighGen Hub · Galaxy Tools · Repository